MICROCONTROLLERS SYSTEMS

II homework (v8)

Mandatory part (6 points):

Connect the circuit (Fig. 1). You will not need all components when programming according to the task, but during the defense the teacher may ask to program some functions additionally.

Complete the following tasks. Anduino libraries are not allowed. Please use plain c for microcontroller programing.

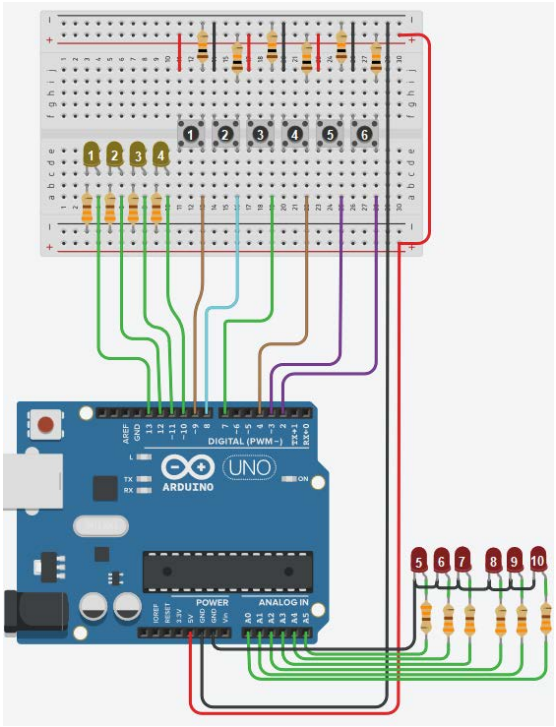


Fig. 1. Circuit no. 1

- The sequence of binary numbers (001, 100, 001, 100, 011, 110, 011, 110, …) should be periodically shown on leds LED8-LED10. The value should change every time the button M5 is pressed.
- The running line to the right should be shown on leds LED1–LED3. The running line should freeze on the last value when the button M4 is pressed. The line should continue to run when the button is pressed second time.

Additional part (4 points):

Connect the circuit (Fig. 2). You will not need all components when programming according to the task, but during the defense the teacher may ask to program some functions additionally.

Complete the following tasks. Anduino libraries are not allowed. Please use plain c for microcontroller programing.

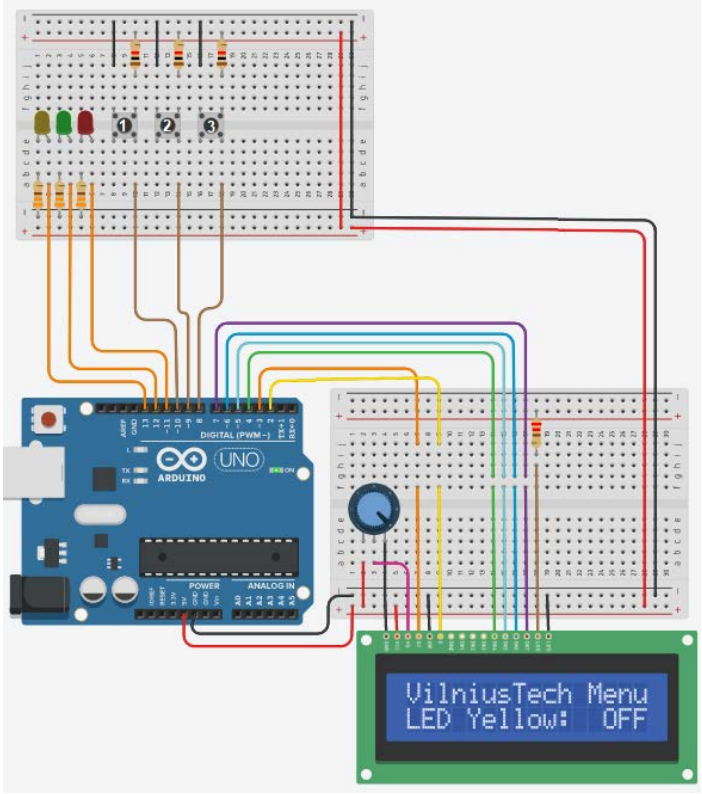


Fig. 2. Circuit no. 2

- Analyze the file "*HW2_main_for_students.c*" which is provided in the moodle system next to the "Additional material" section. The LCD programing functions are presented in this file.
- Create the program that works as follows. Three buttons should control three LEDs. Buttons M1 and M2 are used to select the LED. The M3 button should change the status of the LED (on or off). The text "VilniusTech Menu" should be displayed in the first line of LCD. The second line should consist of two parts. The configurable LED ("LED Yellow: ", "LED Green: " or "LED Red: ") should be shown on the left part. The status of the LED "ON" or "OFF" should be displayed on the right side of the second line.